

# No subset need simultaneously frame an unconditional Schauder frame and its coefficient vectors

Candidate full counterexample to a question in arXiv:2602.21616

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**Status: candidate full counterexample; likely valid; human review needed.**  
 There is a normalized unconditional Schauder frame  $(x_n, y_n)$ , with every  $y_n \neq 0$ , for which no subset  $J$  makes both  $\{x_n/\|x_n\|\}_{n \in J}$  and  $\{\|x_n\|y_n\}_{n \in J}$  frames.

## Source question

Yu considers an unconditional Schauder frame  $\{x_n\}$  with associated coefficient vectors  $\{y_n\}$ , so that

$$z = \sum_n \langle z, y_n \rangle x_n$$

unconditionally for every  $z$  in the Hilbert space. Remark 3.4(b) asks whether there must be a subset  $J$  for which both the normalized primal sequence and the correspondingly rescaled coefficient sequence are frames [1, p. 12]. The exact source text follows.

**Remark 3.4.** (a) In general, one should not expect an arbitrary unconditional Schauder frame for  $H$  to contain a frame for  $H$ . For example, both  $\{n^{-1}e_n\}_{n \in \mathbb{N}}$  and  $\{ne_n\}_{n \in \mathbb{N}}$  are unconditional Schauder frames for  $H$ , but neither of them contains a frame for  $H$ . However, by Lemma 3.2, if a sequence can be rescaled to form a frame by using sufficiently small sequence of scalars, then it contains a frame for  $H$ .

(b) Theorem 3.3 (d) provides a clearer characterization of the scalars chosen to rescale unconditional Schauder frames into a frame than Theorem 2.4 when the notion of an unconditional Schauder frame is formulated as a single sequence. Specifically, for any unconditional Schauder frame  $\{x_n\}_{n \in \mathbb{N}}$  for  $H$  there always exists a subset  $I \subseteq \mathbb{N}$  such that one can always rescale  $\{x_n\}_{n \in \mathbb{N}}$  into a frame by using the rescaling scalars  $c_n = \|x_n\|^{-1}$  if  $n \in I$  and  $c_n = 0$  if  $n \notin I$ . Let  $\{y_n\}_{n \in \mathbb{N}}$  be the sequence of coefficient functionals associated with  $\{x_n\}_{n \in \mathbb{N}}$ , we remark that it remains unclear to us whether there exists some subset  $J \subseteq \mathbb{N}$  such that both  $\{\frac{x_n}{\|x_n\|}\}_{n \in J}$  and  $\{\|x_n\|y_n\}_{n \in J}$  are frames for  $H$ . By Orlicz's Theorem ([47]),  $\{\|x_n\|y_n\}_{n \in J}$  must be a Bessel sequence. However, the existence of a lower frame bound for  $\{\|x_n\|y_n\}_{n \in J}$  remains unknown.  $\diamond$

## Counterexample and proof intuition

**Theorem 1.** *Let  $H = \ell_2(\mathbb{N})$ . There are sequences  $(x_n)$  and  $(y_n)$  in  $H$  such that:*

1.  $\|x_n\| = 1$  and  $y_n \neq 0$  for every  $n$ ;
2.  $z = \sum_n \langle z, y_n \rangle x_n$  unconditionally for every  $z \in H$ ;

3. no subset  $J \subseteq \mathbb{N}$  makes both

$$\left\{ \frac{x_n}{\|x_n\|} \right\}_{n \in J} \quad \text{and} \quad \{\|x_n\|y_n\}_{n \in J}$$

frames for  $H$ .

Consequently the question in Remark 3.4(b) has a negative answer.

**Proof intuition.** Split the reconstruction of the  $k$ th coordinate equally among  $k$  identical copies. The primal sequence then has  $k$  copies of the unit vector  $e_k$ , while every corresponding coefficient vector has size  $1/k$ . If a subset selects  $r_k$  copies from the  $k$ th block, an upper frame bound for the primal sequence forces  $r_k$  to remain bounded. A lower frame bound for the coefficient sequence would instead require  $r_k/k^2$  to stay bounded away from zero. These two requirements are incompatible as  $k \rightarrow \infty$ .

## Proof

Let  $(e_k)_{k \geq 1}$  be the standard orthonormal basis of  $H$ , and use the countable index set

$$\mathcal{I} = \{(k, j) : k \geq 1, 1 \leq j \leq k\}.$$

(Fixing any bijection  $\mathcal{I} \rightarrow \mathbb{N}$  converts the construction into ordinary sequence notation.) For  $(k, j) \in \mathcal{I}$ , set

$$x_{k,j} = e_k, \quad y_{k,j} = \frac{1}{k}e_k. \quad (1)$$

Thus all primal vectors have norm one and all coefficient vectors are nonzero.

If  $z = \sum_{k \geq 1} z_k e_k$ , then the complete  $k$ th block contributes

$$\sum_{j=1}^k \langle z, y_{k,j} \rangle x_{k,j} = \sum_{j=1}^k \frac{z_k}{k} e_k = z_k e_k.$$

This reconstruction is unconditional, not merely conditionally convergent. Indeed, for a finite  $F \subset \mathcal{I}$  put

$$r_k(F) = \#\{j : (k, j) \in F\}, \quad 0 \leq r_k(F) \leq k.$$

The corresponding partial sum is the diagonal contraction

$$S_F z = \sum_{(k,j) \in F} \langle z, y_{k,j} \rangle x_{k,j} = \sum_{k \geq 1} \frac{r_k(F)}{k} z_k e_k. \quad (2)$$

Given  $\varepsilon > 0$ , choose  $K$  so that  $\sum_{k > K} |z_k|^2 < \varepsilon^2$ , and let  $F_0$  contain every index in the first  $K$  blocks. For every finite  $F \supseteq F_0$ , equation (2) gives

$$\|z - S_F z\|^2 = \sum_{k > K} \left| 1 - \frac{r_k(F)}{k} \right|^2 |z_k|^2 \leq \sum_{k > K} |z_k|^2 < \varepsilon^2.$$

Hence the net of finite partial sums converges to  $z$ , which is the unconditional convergence criterion. Therefore (1) is a normalized unconditional Schauder frame with the displayed associated coefficient vectors.

Now let  $J \subseteq \mathcal{I}$  be arbitrary and define its selected multiplicities

$$r_k = \#\{j : (k, j) \in J\}, \quad 0 \leq r_k \leq k.$$

Since  $\|x_{k,j}\| = 1$ , the two frame quadratic forms in the source question are, exactly,

$$\sum_{(k,j) \in J} \left| \left\langle z, \frac{x_{k,j}}{\|x_{k,j}\|} \right\rangle \right|^2 = \sum_{k \geq 1} r_k |z_k|^2, \quad (3)$$

$$\sum_{(k,j) \in J} |\langle z, \|x_{k,j}\| y_{k,j} \rangle|^2 = \sum_{k \geq 1} \frac{r_k}{k^2} |z_k|^2. \quad (4)$$

Suppose first that the sequence in (3) is a frame, and let  $B < \infty$  be an upper frame bound. Testing on  $z = e_k$  yields

$$r_k \leq B \quad (k \geq 1). \quad (5)$$

If the sequence in (4) also had a positive lower frame bound  $A'$ , testing again on  $e_k$  would give

$$A' \leq \frac{r_k}{k^2} \leq \frac{B}{k^2} \quad (k \geq 1),$$

which is impossible as  $k \rightarrow \infty$ . Thus no  $J$  has the two required frame properties, proving the theorem.  $\square$

## Scope and novelty note

The counterexample answers only the simultaneous-subset question in Remark 3.4(b); it does not contradict the paper's theorem that the normalized primal sequence contains a frame subsequence. In fact, choosing one copy of each  $e_k$  gives such a primal orthonormal basis, while its paired coefficient vectors are  $e_k/k$  and have no lower frame bound.

The run's four cheap indexes and bounded searches through 11 August 2026 for the exact remark and its central terms found the source paper but no later resolution. Because the source is recent, novelty remains provisional pending expert literature review.

## References

- [1] P.-T. Yu, *Every semi-normalized unconditional Schauder frame in Hilbert spaces contains a frame*, arXiv:2602.21616v3 (2026).